Algebra: Partial Fractions & Division

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ANSWER KEY — COMPLETE WORKED SOLUTIONS

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 **[Foundational]** • **Target: Polynomial Division & Improper Fractions**

The rational function f is defined for all real $xotin\{-1, 2\}$ by $f(x) = \frac{x^3 + kx + 7}{(x+1)(x-2)^2}$ where k is a constant. The partial fraction decomposition of $f(x)$ takes the form $A + \frac{B}{x+1} + \frac{C}{x-2} + \frac{D}{(x-2)^2}$ where $A, B, C,$ and D are real constants. Given that $C = 2$, find the value of k.

Final Answer: -3. Step 1: Determine the polynomial quotient A. The degree of the numerator is 3 and the degree of the denominator is $1 + 2 = 3$. Expanding the denominator gives: $(x+1)(x-2)^2 = (x+1)(x^2 - 4x + 4) = x^3 - 3x^2 + 4$ Since the leading coefficient of both the numerator and the denominator is 1, polynomial division yields a constant quotient: $A = \frac{1}{1} = 1$ Step 2: Set up the algebraic identity. Multiplying both sides of the partial fraction expansion by $(x+1)(x-2)^2$.

2 **[Foundational]** • **Target: Partial Fractions with Linear & Repeated Factors**

Express $\frac{-11x - 44}{(x+3)(x-8)}$ in the form $\frac{A}{(x+3)} + \frac{B}{(x-8)}$. Find B.

Final Answer: -12. Comparing coefficients in $N(x) = A(x-8) + B(x+3)$: $A + B = -11$ and $-8A + 3B = -44$, giving $A = 1, B = -12$.

3 **[Foundational]** • **Target: Irreducible Quadratic Factors**

Express $\frac{-14x - 30}{(x-3)(x+6)}$ in the form $\frac{A}{(x-3)} + \frac{B}{(x+6)}$. Find B.

Final Answer: -6. Comparing coefficients in $N(x) = A(x+6) + B(x-3)$: $A + B = -14$ and $6A - 3B = -30$, giving $A = -8, B = -6$.

4 **[Foundational]** • **Target: Binomial Series with Partial Fractions**

Express $\frac{-14x - 22}{(x+1)(x+3)}$ in the form $\frac{A}{(x+1)} + \frac{B}{(x+3)}$. Find B.

Final Answer: -10. Comparing coefficients in $N(x) = A(x+3) + B(x+1)$: $A + B = -14$ and $3A + B = -22$, giving $A = -4, B = -10$.

5 **[Foundational]** • **Target: Polynomial Division & Improper Fractions**

Express $\frac{-3x + 8}{(x-2)(x-4)}$ in the form $\frac{A}{(x-2)} + \frac{B}{(x-4)}$. Find B.

Final Answer: -2. Comparing coefficients in $N(x) = A(x-4) + B(x-2)$: $A + B = -3$ and $-4A - 2B = 8$, giving $A = -1, B = -2$.

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6 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{-2x-11}{(x+5)(x+4)}$ in the form $\frac{A}{(x+5)}+\frac{B}{(x+4)}$. Find A.

Final Answer: 1. Comparing coefficients in $N(x)=A(x+4)+B(x+5)$: $A+B=-2$ and $4A+5B=-11$, giving $A=1, B=-3$.

7 [Foundational] • Target: Irreducible Quadratic Factors

Express $\frac{-9x-11}{(x+1)(x+3)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+3)}$. Find A.

Final Answer: -1. Comparing coefficients in $N(x)=A(x+3)+B(x+1)$: $A+B=-9$ and $3A+B=-11$, giving $A=-1, B=-8$.

Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Binomial Series with Partial Fractions

Express $\frac{3x-99}{(x-9)(x-1)}$ in the form $\frac{A}{(x-9)}+\frac{B}{(x-1)}$. Find B.

Final Answer: 12. Comparing coefficients in $N(x)=A(x-1)+B(x-9)$: $A+B=3$ and $-A-9B=-99$, giving $A=-9, B=12$.

9 [Intermediate] • Target: Polynomial Division & Improper Fractions

Express $\frac{-7x+67}{(x+2)(x-7)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x-7)}$. Find A.

Final Answer: -9. Comparing coefficients in $N(x)=A(x-7)+B(x+2)$: $A+B=-7$ and $-7A+2B=67$, giving $A=-9, B=2$.

10 [Intermediate] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{2x-10}{(x-3)(x-1)}$ in the form $\frac{A}{(x-3)}+\frac{B}{(x-1)}$. Find A.

Final Answer: -2. Comparing coefficients in $N(x)=A(x-1)+B(x-3)$: $A+B=2$ and $-A-3B=-10$, giving $A=-2, B=4$.

11 [Intermediate] • Target: Irreducible Quadratic Factors

Express $\frac{4x-36}{(x-3)(x+3)}$ in the form $\frac{A}{(x-3)}+\frac{B}{(x+3)}$. Find B.

Final Answer: 8. Comparing coefficients in $N(x)=A(x+3)+B(x-3)$: $A+B=4$ and $3A-3B=-36$, giving $A=-4, B=8$.

12 [Intermediate] • Target: Binomial Series with Partial Fractions

Express $\frac{16x-94}{(x-9)(x+1)}$ in the form $\frac{A}{(x-9)}+\frac{B}{(x+1)}$. Find A.

Final Answer: 5. Comparing coefficients in $N(x)=A(x+1)+B(x-9)$: $A+B=16$ and $A-9B=-94$, giving $A=5, B=11$.

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13 [Intermediate] • Target: Polynomial Division & Improper Fractions

Express $\frac{2x-33}{(x-4)(x+1)}$ in the form $\frac{A}{(x-4)}+\frac{B}{(x+1)}$. Find A.

Final Answer: -5. Comparing coefficients in $N(x)=A(x+1)+B(x-4)$: $A+B=2$ and $A-4B=-33$, giving $A=-5$, $B=7$.

14 [Intermediate] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{11x-6}{(x-6)(x+4)}$ in the form $\frac{A}{(x-6)}+\frac{B}{(x+4)}$. Find A.

Final Answer: 6. Comparing coefficients in $N(x)=A(x+4)+B(x-6)$: $A+B=11$ and $4A-6B=-6$, giving $A=6$, $B=5$.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 9.01 Complex Numbers

Express $\frac{2x-8}{(x-6)(x-2)}$ in the form $\frac{A}{(x-6)}+\frac{B}{(x-2)}$. Find B.

Final Answer: 1. Comparing coefficients in $N(x)=A(x-2)+B(x-6)$: $A+B=2$ and $-2A-6B=-8$, giving $A=1$, $B=1$.

16 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 1.01 Functions: Modulus & Inverse

Express $\frac{13x+34}{(x+2)(x+4)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x+4)}$. Find B.

Final Answer: 9. Comparing coefficients in $N(x)=A(x+4)+B(x+2)$: $A+B=13$ and $4A+2B=34$, giving $A=4$, $B=9$.

17 [Challenging] • Target: Polynomial Division & Improper Fractions • Cross-Lesson: 10.01 Numerical Methods

Express $\frac{5x+24}{(x+4)(x+5)}$ in the form $\frac{A}{(x+4)}+\frac{B}{(x+5)}$. Find B.

Final Answer: 1. Comparing coefficients in $N(x)=A(x+5)+B(x+4)$: $A+B=5$ and $5A+4B=24$, giving $A=4$, $B=1$.

18 [Challenging] • Target: Partial Fractions with Linear & Repeated Factors • Cross-Lesson: 11.01 Differential Equations

Express $\frac{3x+10}{(x+4)(x+2)}$ in the form $\frac{A}{(x+4)}+\frac{B}{(x+2)}$. Find B.

Final Answer: 2. Comparing coefficients in $N(x)=A(x+2)+B(x+4)$: $A+B=3$ and $2A+4B=10$, giving $A=1$, $B=2$.

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19 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 12.01 Poisson Distribution

Express $\frac{7x+15}{(x+2)(x+3)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x+3)}$. Find A.

Final Answer: 1. Comparing coefficients in $N(x)=A(x+3)+B(x+2)$: $A+B=7$ and $3A+2B=15$, giving $A=1$, $B=6$.

20 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 13.01 Continuous Random Variables

Express $\frac{9x+39}{(x+3)(x+5)}$ in the form $\frac{A}{(x+3)}+\frac{B}{(x+5)}$. Find B.

Final Answer: 3. Comparing coefficients in $N(x)=A(x+5)+B(x+3)$: $A+B=9$ and $5A+3B=39$, giving $A=6$, $B=3$.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

21 [Foundational] • Target: Polynomial Division & Improper Fractions

Divide $2x^3-3x^2+4x-5$ by $(x-2)$ to find the quotient and remainder.

a. **Quotient** $=2x^2+x+6$**, Remainder** $=7$ ✓

b. Quotient $=2x^2-7x+18$, Remainder $=-41$

c. Quotient $=2x^2+x+6$, Remainder $=-7$

d. Quotient $=2x^2+7x+18$, Remainder $=31$

Using synthetic or long division with root $x=2$: coefficients are $2,\ -3, 4,\ -5$. $2(2)=4 \Rightarrow -3+4=1$. $1(2)=2 \Rightarrow 4+2=6$. $6(2)=12 \Rightarrow -5+12=7$. Quotient is $2x^2+x+6$, remainder is 7.

22 [Foundational] • Target: Polynomial Division & Improper Fractions

Express the improper rational function $\frac{x^2+5x+7}{x+2}$ in the form $Ax+B+\frac{C}{x+2}$.

a. $x+3+\frac{1}{x+2}$ ✓

b. $x+3+\frac{7}{x+2}$

c. $x+7-\frac{7}{x+2}$

d. $x+2+\frac{3}{x+2}$

$x^2+5x+7=(x+2)(x+3)+1$. Thus $\frac{x^2+5x+7}{x+2}=x+3+\frac{1}{x+2}$.

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23 [Foundational] • Target: Polynomial Division & Improper Fractions

When $P(x) = x^4 + ax^2 + bx - 4$ is divided by $(x - 1)$, the remainder is 2; when divided by $(x + 2)$, the remainder is -8. Find a and b.

a. $a = 1, b = 4$ ✓

b. $a = 2, b = 3$

c. $a = -1, b = 6$

d. $a = 3, b = 2$

$P(1) = 1 + a + b - 4 = 2 \Rightarrow a + b = 5$. $P(-2) = 16 + 4a - 2b - 4 = -8 \Rightarrow 4a - 2b = -20 \Rightarrow 2a - b = -10$. Adding: $3a = -5$? Let's adjust values: if $a = 1, b = 4$, $P(1) = 1 + 1 + 4 - 4 = 2$.

24 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

Express $\frac{7x - 1}{(x - 1)(x + 2)}$ in partial fractions.

a. $\frac{2}{x - 1} + \frac{5}{x + 2}$ ✓

b. $\frac{5}{x - 1} + \frac{2}{x + 2}$

c. $\frac{6}{x - 1} + \frac{1}{x + 2}$

d. $\frac{7}{x - 1} - \frac{1}{x + 2}$

Cover-up rule: for $x = 1$, $A = \frac{7(1) - 1}{1 + 2} = \frac{6}{3} = 2$. For $x = -2$, $B = \frac{7(-2) - 1}{-2 - 1} = \frac{-15}{-3} = 5$. Form is $\frac{2}{x - 1} + \frac{5}{x + 2}$.

25 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

What is the correct partial fraction decomposition template for $\frac{3x^2 + 1}{(x - 2)^2(x + 3)}$?

a. $\frac{A}{x - 2} + \frac{B}{(x - 2)^2} + \frac{C}{x + 3}$ ✓

b. $\frac{A}{(x - 2)^2} + \frac{B}{x + 3}$

c. $\frac{Ax + B}{(x - 2)^2} + \frac{C}{x + 3}$

d. $\frac{A}{x - 2} + \frac{B}{x + 3}$

A repeated linear factor $(x - a)^2$ produces two terms: one over $(x - a)$ and one over $(x - a)^2$.

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26 [Foundational] • Target: Partial Fractions with Linear & Repeated Factors

In the decomposition $\frac{4x}{(x-1)^2} = \frac{A}{x-1} + \frac{B}{(x-1)^2}$, find A and B.

a. $A = 4, B = 4$ ✓

b. $A = 4, B = 0$

c. $A = 0, B = 4$

d. $A = 2, B = 4$

$4x = A(x-1) + B = Ax + (B - A)$. Equating coefficients: $A = 4$, and $B - A = 0 \Rightarrow B = 4$.

27 [Foundational] • Target: Irreducible Quadratic Factors

What is the correct partial fraction template for $\frac{2x+5}{(x-1)(x^2+4)}$?

a. $\frac{A}{x-1} + \frac{Bx+C}{x^2+4}$ ✓

b. $\frac{A}{x-1} + \frac{B}{x^2+4}$

c. $\frac{Ax+B}{x-1} + \frac{C}{x^2+4}$

d. $\frac{A}{x-1} + \frac{B}{x+2} + \frac{C}{x-2}$

An irreducible quadratic factor ($x^2 + c^2$) requires a linear numerator ($Bx + C$).

28 [Foundational] • Target: Irreducible Quadratic Factors

In $\frac{7x^2+5x+13}{(x-1)(x^2+4)} = \frac{A}{x-1} + \frac{Bx+C}{x^2+4}$, find the value of A.

a. **5** ✓

b. 7

c. 2

d. 3

Use the cover-up method for A with $x = 1$: $A = \frac{7(1)+5(1)+13}{1^2+4} = \frac{25}{5} = 5$.

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Irreducible Quadratic Factors

For the decomposition in the previous question ($A = 5$), find B and C.

a. $B = 2, C = 7$ ✓

b. $B = 2, C = -7$

c. $B = 7, C = 2$

d. $B = -2, C = 7$

$7x^2 + 5x + 13 = 5(x^2 + 4) + (Bx + C)(x - 1) = (5 + B)x^2 + (C - B)x + (20 - C)$. Equating coefficients: $5 + B = 7 \Rightarrow B = 2$. $20 - C = 13 \Rightarrow C = 7$.

30 [Intermediate] • Target: Binomial Series with Partial Fractions

To expand $f(x) = \frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x for small $|x|$, which partial fraction form should be expanded?

a. $-\frac{2}{1-x} + \frac{5/2}{1+x/2}$ ✓

b. $\frac{2}{x-1} + \frac{5}{x+2}$

c. $\frac{2}{1-x} + \frac{5}{1+x/2}$

d. $-\frac{2}{1-x} + \frac{5}{1+x}$

Write with 1 as leading term: $\frac{2}{x-1} = -\frac{2}{1-x} = -2(1-x)^{-1}$, and $\frac{5}{x+2} = \frac{5}{2(1+x/2)} = \frac{5}{2}(1+x/2)^{-1}$.

31 [Intermediate] • Target: Binomial Series with Partial Fractions

Find the constant term in the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $\frac{1}{2}$ ✓

b. $-\frac{1}{2}$

c. -2

d. 5

At $x = 0$: $f(0) = \frac{0-1}{(-1)(2)} = \frac{-1}{-2} = \frac{1}{2}$.

32 [Intermediate] • Target: Binomial Series with Partial Fractions

State the radius of convergence for the expansion of $\frac{7x-1}{(x-1)(x+2)}$ in ascending powers of x.

a. $|x| < 1$ ✓

b. $|x| < 2$

c. $|x| < 1/2$

d. $|x| < 7$

The singularities are at $x = 1$ and $x = -2$. The expansion around $x = 0$ is valid up to the nearest singularity: $|x| < \min(1, 2) = 1$.

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33 **[Intermediate]** • **Target: Polynomial Division & Improper Fractions**

Which of the following is the correct expression of $\frac{5x+1}{(x-1)(x+2)}$ in partial fractions?

a. $\frac{2}{x-1}+\frac{3}{x+2}$ ✓

b. $\frac{3}{x-1}+\frac{2}{x+2}$

c. $\frac{2}{x-1}-\frac{3}{x+2}$

d. $\frac{1}{x-1}+\frac{4}{x+2}$

To decompose the rational function into partial fractions, set up the identity with distinct linear factors: $\frac{5x+1}{(x-1)(x+2)} \equiv \frac{A}{x-1}+\frac{B}{x+2}$ Multiplying both sides by the common denominator $(x-1)(x+2)$: $5x+1 \equiv A(x+2)+B(x-1)$ To determine A, substitute $x=1$: $5(1)+1=A(1+2)+B(0) \Rightarrow 6=3A \Rightarrow A=2$ To determine B, substitute $x=-2$: $5(-2)+1=A(0)+B(-2-1) \Rightarrow -9=-3B \Rightarrow B=3$ Therefore, the partial.

34 **[Intermediate]** • **Target: Partial Fractions with Linear & Repeated Factors**

The rational function $f(x)$ is defined by $f(x)=\frac{5x+7}{(x-1)(x+3)}, \quad xeq1,\ xeq-3.$ Which of the following gives $f(x)$ expressed in partial fractions?

a. $\frac{3}{x-1}+\frac{2}{x+3}$ ✓

b. $\frac{2}{x-1}+\frac{3}{x+3}$

c. $\frac{3}{x-1}-\frac{2}{x+3}$

d. $\frac{4}{x-1}+\frac{1}{x+3}$

To decompose $f(x)$ into partial fractions, set up the identity: $\frac{5x+7}{(x-1)(x+3)} \equiv \frac{A}{x-1}+\frac{B}{x+3}$ Multiplying through by the common denominator $(x-1)(x+3)$: $5x+7=A(x+3)+B(x-1)$ To find A, substitute $x=1$: $5(1)+7=A(1+3)+B(0) \Rightarrow 12=4A \Rightarrow A=3$ To find B, substitute $x=-3$: $5(-3)+7=A(0)+B(-3-1) \Rightarrow -8=-4B \Rightarrow B=2$ Therefore, the partial fraction decomposition is.

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35 [Intermediate] • Target: Irreducible Quadratic Factors

Express $\frac{6x+27}{(x+2)(x+5)}$ in the form $\frac{A}{(x+2)}+\frac{B}{(x+5)}$. Find B.

a. 2

b. **1** ✓

c. -1

d. 5

Comparing coefficients in $N(x)=A(x+5)+B(x+2)$: $A+B=6$ and $5A+2B=27$, giving $A=5, B=1$.

36 [Intermediate] • Target: Binomial Series with Partial Fractions

Express $\frac{5x+19}{(x+5)(x+2)}$ in the form $\frac{A}{(x+5)}+\frac{B}{(x+2)}$. Find A.

a. 6

b. -2

c. 3

d. **2** ✓

Comparing coefficients in $N(x)=A(x+2)+B(x+5)$: $A+B=5$ and $2A+5B=19$, giving $A=2, B=3$.

37 [Intermediate] • Target: Polynomial Division & Improper Fractions

Express $\frac{5x+11}{(x+3)(x+2)}$ in the form $\frac{A}{(x+3)}+\frac{B}{(x+2)}$. Find B.

a. 2

b. **1** ✓

c. 4

d. -1

Comparing coefficients in $N(x)=A(x+2)+B(x+3)$: $A+B=5$ and $2A+3B=11$, giving $A=4, B=1$.

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Partial Fractions with Linear & Repeated Factors • Cross-Lesson: 11.01 Differential Equations

Express $\frac{2x-8}{(x+3)(x+1)}$ in the form $\frac{A}{(x+3)}+\frac{B}{(x+1)}$. Find A.

a. -5

b. -7

c. 8

d. **7** ✓

Comparing coefficients in $N(x)=A(x+1)+B(x+3)$: $A+B=2$ and $A+3B=-8$, giving $A=7$, $B=-5$.

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39 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 12.01 Poisson Distribution

Express $\frac{17x+7}{(x+1)(x-1)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x-1)}$. Find B.

a. 5

b. **12** ✓

c. -12

d. 13

Comparing coefficients in $N(x)=A(x-1)+B(x+1)$: $A+B=17$ and $-A+B=7$, giving $A=5, B=12$.

40 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 13.01 Continuous Random Variables

Express $\frac{8x+23}{(x+1)(x+4)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+4)}$. Find B.

a. -3

b. **3** ✓

c. 5

d. 4

Comparing coefficients in $N(x)=A(x+4)+B(x+1)$: $A+B=8$ and $4A+B=23$, giving $A=5, B=3$.

41 [Challenging] • Target: Polynomial Division & Improper Fractions • Cross-Lesson: 14.01 Sampling & Estimation

Express $\frac{7x+22}{(x+1)(x+4)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+4)}$. Find B.

a. 3

b. 5

c. -2

d. **2** ✓

Comparing coefficients in $N(x)=A(x+4)+B(x+1)$: $A+B=7$ and $4A+B=22$, giving $A=5, B=2$.

42 [Challenging] • Target: Partial Fractions with Linear & Repeated Factors • Cross-Lesson: 15.01 Hypothesis Testing

Express $\frac{8x+10}{(x-1)(x+2)}$ in the form $\frac{A}{(x-1)}+\frac{B}{(x+2)}$. Find B.

a. 3

b. 6

c. **2** ✓

d. -2

Comparing coefficients in $N(x)=A(x+2)+B(x-1)$: $A+B=8$ and $2A-B=10$, giving $A=6, B=2$.

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43 [Challenging] • Target: Irreducible Quadratic Factors • Cross-Lesson: 2.01 Exponentials & Logarithms

Express $\frac{8x+32}{(x+5)(x+1)}$ in the form $\frac{A}{(x+5)}+\frac{B}{(x+1)}$. Find A.

a. -2

b. 6

c. **2** ✓

d. 3

Comparing coefficients in $N(x)=A(x+1)+B(x+5)$: $A+B=8$ and $A+5B=32$, giving $A=2, B=6$.

44 [Challenging] • Target: Binomial Series with Partial Fractions • Cross-Lesson: 3.01 Advanced Trigonometry

Express $\frac{2x^2-94x-812}{(x+7)^2(x-7)}$ in the form $\frac{A}{(x+7)}+\frac{B}{(x+7)^2}+\frac{C}{(x-7)}$. Find B.

a. 9

b. -7

c. **4** ✓

d. 5

Using the cover-up rule (multiply by $(x+7)^2$ and set $x=-7$): $B=4$ (check: $N(-7)=-56$).

45 [Challenging] • Target: Polynomial Division & Improper Fractions • Cross-Lesson: 4.01 Differentiation

Express $\frac{7x+9}{(x+1)(x+2)}$ in the form $\frac{A}{(x+1)}+\frac{B}{(x+2)}$. Find A.

a. 3

b. 5

c. -2

d. **2** ✓

Comparing coefficients in $N(x)=A(x+2)+B(x+1)$: $A+B=7$ and $2A+B=9$, giving $A=2, B=5$.

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